

Distance-Preserving Multiset Hashes: Separating Agreement Estimation from Agreement Security in Byzantine Consensus

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Abstract—Byzantine fault tolerant (BFT) protocols compare validator state with collision-resistant hashes, and collision resistance destroys distance: validators that agree on 19 of 20 transactions produce unrelated hashes, indistinguishable from validators that share nothing. We argue that agreement *estimation* and agreement *security* are different jobs that deserve different objects, and we show that one linear-algebraic object can serve both. A distance-preserving multiset hash (DPMH) maps each transaction to a zero-mean vector in $\{-1, +1\}^n$ and digests a set by coordinate-wise integer summation, so that $\|\Delta D\|^2/n$ is an unbiased estimator of the multiset symmetric difference with relative error at most $\sqrt{2/n}$, while equality of the mod- q digest is a collision-binding property whose violation reduces to a short-integer-solution problem over a ± 1 matrix. We prove that no distance-preserving hash can make *proximity* authoritative, which forces an architecture: distance may gate scheduling, synchronization, and optimism, never safety; and we prove a composition theorem showing that any BFT protocol whose commit predicate is a quorum of signatures retains its safety unchanged under such an estimation layer. On this foundation we rebuild Proxima, a two-phase BFT protocol with a speculative fast path whose trigger is honest-sufficient (a quorum of signed commitments, not a variance statistic an adversary can perturb), a Byzantine-robust reference (coordinate-wise median), straggler synchronization by estimate-then-decode set reconciliation, $O(\log H)$ fork localization, partition healing at cost proportional to divergence, and a cross-shard *conservation invariant* that verifies sharded settlement with volume-independent metadata and localizes faulty corridors in $O(f \log C)$ signed probes. Simulations confirm every closed form and show the protocol consequences: a single in-cluster Byzantine validator disables the prior fast path but not ours; a Byzantine aggregator excludes every honest validator under the prior reference and none under ours; and message complexity stays 1.5 to $2\times$ below HotStuff-2 at $N \leq 3000$ while the bandwidth premium of kilobyte digests inverts in our favor beyond $N \approx 2000$.

Index Terms—Byzantine fault tolerance, consensus, multiset hashing, sketching, set reconciliation, sharding, blockchain.

I. INTRODUCTION

BYZANTINE fault tolerant consensus has been built on collision-resistant hashing since PBFT [2]. Collision resistance is the right property for preventing forgery and

exactly the wrong property for measuring agreement: when two validators hash their transaction sets and the hashes differ, the protocol learns one bit, not identical, whether the sets differ by one transaction or by half. Three structural costs follow. Validators must synchronize state before they can vote; agreement quality is unmeasurable until votes are counted, so protocols such as HotStuff [3] run every voting round even under unanimous agreement; and hierarchical designs need committees large enough to run independent BFT, which is why Ethereum’s attestation committees hold 128 validators [4] and why cross-shard transactions settle through per-transaction coordination.

The obvious repair, replacing the hash with a similarity-preserving digest, runs into an equally obvious objection: a digest that maps nearby sets to nearby values cannot be collision-resistant, so nothing it says can be trusted against an adversary. Our starting point is that this objection, made precise, is not a refutation but a design rule. We prove that for *any* distance-preserving multiset hash, digest proximity is unbindable: an adversary can always exhibit sets within any threshold that differ in ways a protocol cares about, because the distance property itself guarantees it (Theorem 1). The conclusion is not to abandon estimation but to route it away from safety. Estimation gates *scheduling*: when to propose, whom to synchronize, when to take an optimistic path. Security remains what it always was: a quorum of signatures on a collision-resistant hash. We formalize this separation and prove it composes: any BFT protocol whose commit predicate is a signature quorum retains its safety argument verbatim when an estimation layer is added (Theorem 2).

What makes the separation productive rather than merely safe is that a single object serves both sides of it. Our distance-preserving multiset hash (DPMH) digests a transaction multiset by summing per-transaction Rademacher vectors, giving three properties no collision-resistant hash offers: the squared digest distance is an unbiased estimator of the multiset symmetric difference; digests add and subtract, so groups, histories, and in-flight sets have exact one-object summaries; and the same additive structure over a modulus is a homomorphic accumulator whose equality is collision-binding under a short-integer-solution (SIS) assumption, the algebra behind Ajtai hashing [5] and deployed in Facebook’s LtHash [6]. Estimation uses the geometry; security uses only equality; the impossibility theorem says this division of labor is forced, and the construction says it is cheap: one SHA-512 and n integer additions per transaction, one kilobyte on the wire.

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Code and reproduction artifacts: <https://github.com/RyanMercier/Proxima>. An earlier version of this work appeared at the 8th International Congress on Blockchain and Applications [1]; this article replaces that version’s primitive with a zero-mean construction, adds the binding accumulator, the impossibility and composition theorems, robust reference selection, readiness sensing, exact set reconciliation, and conservation sharding, and evaluates against HotStuff-2 and Kauri.

We rebuild the Proxima protocol of our conference paper [1] on this foundation and find that every protocol mechanism sharpens:

- **Unbiased estimation (Sec. IV).** The conference version used nonnegative coordinates, whose distance signal depends on the direction of the difference: substituting j transactions reads at a fraction of its true size and can hide below honest stragglers. Zero-mean coordinates measure exactly the symmetric difference (Lemma 1), the clustering threshold becomes the closed form $\tau^2 = 3n$ with honest-exclusion probability $e^{-n/8}$, and Monte Carlo calibration is demoted to a validation check.
- **Honest-sufficient fast path (Sec. VI).** Speculative signed commitments collected in round one form a finality certificate identical to the slow-path object, so the fast path is covered by the safety theorem verbatim; and because the trigger counts signatures rather than testing a variance statistic, one in-cluster Byzantine validator can no longer disable it. In simulation the prior trigger’s fast-path rate falls from 100% to 0% at a single mimicking adversary; ours is unaffected (Fig. 4).
- **Byzantine-robust reference (Sec. VI).** The clustering reference becomes the coordinate-wise median of the submitted digests, verifiable by any validator, with breakdown point $1/2$ per coordinate. Under a coordinated drag attack with a Byzantine aggregator, the prior aggregator-chosen reference excludes 100% of honest validators; the median excludes none up to 45% corruption (Fig. 3).
- **Readiness sensing (Sec. VI).** Digests piggybacked on gossip let the proposer measure convergence and propose when a quorum already matches, converting fast-path engagement from weather, $(1 - p_{\text{miss}})^N$, into a controlled quantity: at 80% initial miss rate, fixed-interval proposal never engages the fast path and sensing engages it in every trial at a mean delay of 3.4 gossip ticks (Fig. 4).
- **Estimate-then-decode synchronization (Sec. VII).** The digest distance sizes an invertible Bloom lookup table [7] that recovers exactly the differing transactions, replacing the conference version’s Bloom filters and their false-positive liveness channel. The same two-syndrome pattern heals partitions at cost proportional to divergence and, with cumulative accumulators, localizes chain forks in exactly $\log_2 H + 1$ probes.
- **Conservation sharding (Sec. VIII).** Cross-shard settlement becomes double-entry bookkeeping in vector space: outbound and inbound corridor accumulators must sum to equality, a collision-binding invariant checked with $O(S)$ kilobyte objects per block regardless of volume. Violations are localized to faulty corridors by group testing in $O(f \log C)$ signed probes and resolved exactly by corridor syndromes: in a 16-shard run with 9 injected faults the invariant’s violation norm estimated 8.9 dangling effects, aging alarms and bisection named exactly the 2 faulty corridors in 15 probes, and the decode recovered every dropped and minted transaction.

Sections II through III position and set up the work; Sections IV and V define the primitive and its security

envelope; Sections VI through VIII build the protocol layer; and Section IX evaluates every claim against HotStuff, HotStuff-2 [8], Kauri [9], and PBFT, with the honest costs (a bandwidth premium at small N , a simulation-only Kauri model, open cryptanalysis at exact parameters) stated where they arise.

II. RELATED WORK

BFT consensus and speculation. PBFT [2] established practical BFT at $O(N^2)$ messages; HotStuff [3] reached linearity with leader-based aggregation and threshold signatures, HotStuff-2 [8] removed one of its three phases, and Tendermint [10] offers a similar structure. All compare state by collision-resistant hash and synchronize fully before voting. Our fast path descends from speculative execution in Zyzzyva [11] and SBFT [12], where optimistically collected signatures finalize early when a large enough quorum already agrees; our contribution is not the speculation but the estimation layer that decides when to attempt it and the theorem that such layers cannot harm safety.

Hierarchical and DAG designs. Kauri [9] pipelines tree-based dissemination and aggregation over HotStuff but keeps per-group, hash-based voting, hence groups large enough for Byzantine tolerance; Ethereum’s 128-validator committees [4] exist for the same reason. Distance filtering removes the per-group vote, so our tree runs on groups of ten. Narwhal/Tusk [13] and Bullshark [14] decouple data availability from ordering over partial views; digests could measure vertex agreement there, which we leave open.

Multiset hashing and sketches. Additive multiset hashes go back to AdHash [15], broken into enormous moduli by Wagner’s generalized birthday attack [16], and were formalized by Clarke et al. [17]; the vectorized form we use is deployed as LtHash [6], whose SIS-style analysis [5] we inherit. The estimation side is the AMS/Tug-of-War sketch [18] specialized to multiset indicator vectors. What we add is the observation that one object carries both properties at once, a quantitative adversarial-deflation analysis in the public-hash setting with the two-domain design it forces, and the consensus layer built on top. Hardt and Woodruff [19] proved linear sketches are not robust to adaptive adversaries; our impossibility theorem is the public-coin multiset-hash analogue, and our composition theorem is the architectural answer. Locality-sensitive hashing [20]–[23] solves nearest neighbors, not summable state comparison; we do not use it.

Set reconciliation. Exact reconciliation descends from characteristic polynomials [24] through invertible Bloom lookup tables [7], Graphene [25], minisketch as deployed in Bitcoin’s Eray [26], and rateless IBLTs [27]. Eray is the existence proof that set reconciliation belongs inside blockchain protocols; we generalize its role from transaction relay to consensus synchronization, fork healing, and cross-shard resolution, with the digest supplying the size estimate that the decoder needs.

Robust aggregation. Coordinate-wise medians and trimmed means with Byzantine breakdown guarantees come from robust distributed learning [28], [29]; we import them as reference selection for clustering, which to our knowledge is new in consensus.

Sharding. Ethereum moved to data-availability sharding (Danksharding [30]); NEAR’s Nightshade [31] settles with receipts whose volume tracks transaction volume. Our conservation invariant is complementary to both: constant-size settlement verification with fraud localization, not an execution-ordering mechanism.

III. SYSTEM MODEL

We consider N validators, at most $f < N/3$ Byzantine, behaving arbitrarily and coordinating; honest validators follow the protocol. We assume partial synchrony [32]: after an unknown global stabilization time, messages between honest validators arrive within a known bound. Each validator holds a BLS [33] key pair with public keys known to all; signatures aggregate (N signatures into one 96-byte aggregate), and we assume CDH hardness in BLS12-381 [34]. Communication runs through an aggregator (flat mode) or a tree of relays (tree mode); the role rotates every round, and a Byzantine aggregator can suppress messages, which costs liveness, but cannot forge signatures. The adversary is static, corrupting validators before the protocol begins; it observes the full network and may grind transactions (Section V). SHA-512 is modeled as a random oracle. Honest validators may receive transactions with delay relative to the proposer; p_{miss} denotes the per-validator probability of an incomplete view at proposal time, and an incomplete honest view misses at most $k_{\text{max}} = 2$ transactions. Blocks are sets: multiplicity within a block is bounded by one and enforced at validation.

IV. THE PRIMITIVE

A. Construction

For round (height) h , the per-transaction vector is

$$v_h(tx) = \text{sgn}(H(\text{dom} \parallel h \parallel tx)) \in \{-1, +1\}^n, \quad (1)$$

where H is SHA-512, sgn maps each of the first n output bits b to $2b - 1$, and dom is a domain-separation tag. For $n = 512$ one hash invocation suffices. The digest of a multiset T is the coordinate-wise integer sum

$$D_h(T) = \sum_{tx \in T} v_h(tx), \quad (2)$$

stored as n signed 16-bit words: each transaction moves each coordinate by exactly one, so for any block with $|T| \leq 32767$ the mod- 2^{16} encoding equals the integer digest with no wraparound. Homomorphism is immediate in both directions, $D(A \oplus B) = D(A) + D(B)$ and $D(A) - D(B)$ is meaningful, and group summaries travel as exact integer pairs (sum of digests, total weight) so that no floating point ever enters the protocol, eliminating cross-platform float nondeterminism as a bug class. Computation is one SHA-512 plus n integer additions per transaction and one $O(n)$ dot product per distance.

Two domains coexist, and Section V shows why both are needed. *Round digests* are salted by height ($\text{dom} = \text{PXM}/\text{round}$, $q = 2^{16}$, $n = 512$) and feed clustering, sensing, and synchronization. *Ledger accumulators* use a fixed salt ($\text{dom} = \text{PXM}/\text{ledger}$, $q = 2^{32}$) and feed cumulative chain digests $P[b] = \sum_{b' \leq b} D(\text{block}_{b'})$, conservation invariants, and

TABLE I
PARAMETER PROFILES. RSE IS THE RELATIVE STANDARD ERROR OF THE SYMMETRIC-DIFFERENCE ESTIMATE; EXCLUSION IS THE PROBABILITY THAT AN HONEST VALIDATOR WITH $d \leq 2$ EXCEEDS $\tau^2 = 3n$.

profile	n	q	wire	RSE	honest excl.
round	512	2^{16}	1024 B	$\leq 6.3\%$	e^{-64}
light	128	2^{16}	256 B	$\leq 12.5\%$	e^{-16}
ledger	512	2^{32}	2048 B	exact [†]	n/a

[†]Distance on accumulator differences is exact below $q/2 = 2^{31}$ divergent transactions; equality binding is unaffected by saturation.

fork localization, where equality must be binding and algebra must compose across heights. Table I lists the profiles.

B. Unbiased Distance Estimation

Throughout this subsection the two multisets are fixed before the oracle is queried; the adversarial regime is Section V.

Lemma 1 (Unbiased symmetric-difference estimator): Let $d = |A \triangle B|$ and $\Delta D = D(A) - D(B)$. Then ΔD is distributed as a sum of d i.i.d. uniform $\{-1, +1\}^n$ vectors, and

$$\mathbb{E}[\|\Delta D\|^2] = nd, \quad \hat{d} := \|\Delta D\|^2/n \text{ is unbiased}, \quad (3)$$

with $\text{Var}(\hat{d}) = 2d(d-1)/n$, hence relative standard error at most $\sqrt{2/n}$ (6.25% at $n = 512$).

Proof: Vectors of elements in $A \cap B$ cancel in ΔD ; the remaining d vectors appear with signs ± 1 , which the symmetry of the Rademacher distribution absorbs. Per coordinate, $S = \sum_{i=1}^d \varepsilon_i$ with ε_i i.i.d. Rademacher has $\mathbb{E}[S] = 0$, $\mathbb{E}[S^2] = d$, and $\mathbb{E}[S^4] = 3d^2 - 2d$. Summing $\mathbb{E}[S^2]$ over n independent coordinates gives $\mathbb{E}\|\Delta D\|^2 = nd$; the variance of $\|\Delta D\|^2$ is $n(\mathbb{E}[S^4] - \mathbb{E}[S^2]^2) = n(2d^2 - 2d)$, and dividing by n^2 yields the claim. ■

Two consequences shape the protocol. First, direction independence: substituting j transactions is a symmetric difference of $2j$ and is measured as such, where the non-negative coordinates of the conference version [1] let substitutions partially cancel and read below honest stragglers. Second, exact small- d laws that serve as implementation test vectors: at $d = 1$, $\|\Delta D\|^2 = n$ deterministically (a singleton difference is certified by the exact value n , with zero variance); at $d = 2$, $\|\Delta D\|^2 \sim 4 \text{Bin}(n, 1/2)$; at $d = 3$, $\|\Delta D\|^2 \sim n + 8 \text{Bin}(n, 1/4)$. Coordinates are independent, so Hoeffding also gives the crude tail $\Pr[|\|\Delta D\|^2 - nd| \geq nt] \leq 2 \exp(-2nt^2/d^4)$ for larger d .

C. Closed-Form Threshold

An honest validator with an incomplete view misses at most $k_{\text{max}} = 2$ transactions, so its digest sits at squared distance concentrated on $nd \leq 2n$ from the full-state digest. We therefore cluster at

$$\tau^2 = 3n. \quad (4)$$

Honest exclusion at $d = 2$ requires $4 \text{Bin}(n, 1/2) > 3n$, i.e., a binomial deviation of $n/8$ above its mean, of probability at most $e^{-n/8}$ ($e^{-64} \approx 1.6 \times 10^{-28}$ at $n = 512$); at $d = 1$ exclusion

is impossible since $n < 3n$. A fabrication at $d = 4$ sits 4.6 standard deviations above the threshold. The value $d = 3$ is genuinely ambiguous and is resolved by synchronization plus the signature phase, never by the estimator; this ambiguity is inherent to any finite-variance estimator and costs at most one round, never safety. The conference version calibrated its threshold by Monte Carlo; here the threshold is analytic and the simulation is only a validation check (Section IX).

V. SECURITY OF THE PRIMITIVE

A. Equality Binding

A collision for the accumulator is a pair of distinct bounded-multiplicity multisets with equal digests mod q , equivalently a nonzero integer vector z with bounded entries and bounded support such that $Vz \equiv 0 \pmod{q}$, where the columns of V are the oracle-derived ± 1 vectors of the ground set. This is a short-integer-solution instance with a ± 1 matrix, the algebra of Ajtai hashing [5] and of LtHash [6]. Collisions exist by pigeon-hole, as they must for any compressing homomorphic hash; security is the claim that finding one is infeasible. Wagner’s k -tree attack [16] over $\mathbb{Z}_q^n$ must cancel $n \log_2 q = 8192$ constraint bits and produces solutions of Hamming weight 2^a using lists of cost roughly $2^{8192/(a+1)}$; the block-size cap $|T| \leq 2^{15}$ forces $a \leq 15$ and per-list cost 2^{512} . Lattice attacks must find vectors in the q -ary kernel lattice of V far shorter than the Gaussian heuristic at these parameters. We state binding as a reduction shape with these first-cut estimates and inherit the parameter analysis of LtHash [6] at the same $n \log q$ budget; a dedicated cryptanalysis at our exact parameters is open (Section X), and we claim no concrete security level without it.

B. Adversarial Deflation and the Two Domains

Distance is an estimator, not a commitment, and a grinding adversary can make a large true difference look smaller. The cheapest strategy is near-antipodal pairs: for a target $v = v(tx)$, find tx' whose vector agrees with $-v$ on all but a coordinates, so the pair contributes $4a$ to the squared distance instead of the honest expectation $2n$. Agreement counts are $\text{Bin}(n, 1/2)$, so a pair achieving $a \leq n/4$ (factor-two deflation) occurs with probability $e^{-n/8} = 2^{-92}$ at $n = 512$, and a birthday search over M candidates yields about $M^2 2^{-93}$ usable pairs: one pair per 2^{47} hash-and-sign candidates, each of which costs the adversary a signable transaction. Factor-four deflation costs 2^{104} candidates per pair. Deflation is therefore bounded to constant factors at exponential grinding cost, and per-height salting of the round domain forces any precomputed table to be rebuilt every block, converting a one-time investment into an online per-block cost.

Salting, however, destroys cross-height algebra, which the accumulators need. Hence the two domains of Section IV: the salted round domain feeds thresholds, where deflation matters and no cross-height algebra is needed; the fixed ledger domain feeds equality and conservation checks, which rest on binding, not on distance, so deflation is irrelevant there. Each domain carries exactly the property it needs and no more.

C. Proximity Is Unbindable

Theorem 1 (Proximity non-bindability): Let D be any multiset hash with the distance-preservation property that $|A \triangle B| \leq d_0$ implies $\|D(A) - D(B)\|$ small (below any protocol threshold τ) with non-negligible probability. Then there is no security notion under which digest proximity certifies set proximity against adversarially chosen inputs: for every threshold τ , the adversary can exhibit pairs of sets within τ that differ in protocol-relevant ways. Consequently, in any protocol composed with D , proximity may gate scheduling, synchronization, and optimistic paths, but never a safety predicate.

Proof: For small differences the claim is the property itself: replace one transaction by an adversarial one; distance preservation puts the resulting set within τ for any usable τ , and the sets differ in exactly the transaction the adversary chose. For larger differences, grinding (Section V-B) extends the reach at quantifiable cost, without ever needing a collision. Any protocol that commits state on a proximity predicate therefore commits, with adversarial help, to a set other than intended, violating safety by construction. The only predicates immune to this argument are those independent of D ’s geometry, i.e., exact equality (Section V-A) and external authentication such as signatures. ■

Theorem 1 is the public-coin, multiset-hash analogue of the Hardt-Woodruff theorem that linear sketches are not robust to adaptive inputs [19]. The resolutions differ: rather than seeking a robust sketch, we route all safety through signatures on collision-resistant hashes and let the estimator fail soft. The conference version stated this honesty informally (“Phase 1 contributes no cryptographic security”); the theorem makes the architecture forced rather than stylistic.

VI. ESTIMATION-AUGMENTED CONSENSUS

Definition 1 (Estimation augmentation): Let Π be a BFT protocol whose commit predicate is “at least $2N/3$ valid signatures on $H(B)$ ” for a collision-resistant H . An *estimation augmentation* $\Pi + E$ adds messages and rules that may depend on DPMH digests, but whose outputs gate only scheduling (when to propose, when to attempt a fast path), synchronization (which transactions to push to whom), and reporting. The commit predicate is unchanged.

Theorem 2 (Safety preservation): $\Pi + E$ satisfies the same safety property as Π against the same adversary class.

Proof: Estimation messages are auxiliary: digests are computable from public data and require no secrets. Given any adversary $\mathcal{A}$ against $\Pi + E$, construct $\mathcal{A}'$ against Π that internally simulates the estimation layer, forwarding to Π exactly the commit messages $\mathcal{A}$ produces. The executions agree on all commit messages, so any safety violation in $\Pi + E$ is one in Π . Π ’s safety argument, quorum intersection on signed $H(B)$, never invokes any property of D . ■

Corollary 1 (Speculative fast path): In Phase 1 each validator sends (D_i, σ_i) with σ_i a BLS signature on $(h, H(B_{\text{prop}}))$, attached iff its local state matches the proposal. If the aggregator collects $2N/3$ matching signatures in round one, it aggregates them and multicasts a finality certificate identical to the object

the signature phase produces (96-byte aggregate plus $N/8$ -byte signer bitmap). The fast path is a scheduling event under Definition 1, covered by Theorem 2; an aggregator cannot fabricate a certificate and can at most withhold one, a liveness effect.

The corollary repairs two defects of the conference version at once. Its fast-path certificate was issued from digest agreement alone, backed by no signatures, which Theorem 1 now rules out on principle. And its trigger tested a global statistic, near-zero cluster variance, which a single Byzantine validator inside the threshold could perturb forever, disabling the fast path for free. Counting signatures is *honest-sufficient*: below the corruption bound the honest $2N/3$ match regardless of what the adversary submits. We state the design rule explicitly because it generalizes: *optimistic triggers must fire on honest-sufficient conditions, never on global statistics the adversary can perturb*.

Robust reference. The clustering reference in the conference version was the aggregator’s own state, a single manipulation point: a Byzantine aggregator could place the reference anywhere and exclude every honest validator. The reference is now the coordinate-wise median of the submitted, signed Phase 1 digests, computable and verifiable by any validator, with breakdown point $1/2$ per coordinate imported from robust distributed learning [28], [29]. A coalition below one half cannot move any coordinate of the reference outside the range of honest values, and the aggregator retains no discretion.

Readiness sensing. Because digests ride on gossip, the proposer can watch the digest cloud converge and propose when the number of exactly-matching validators reaches the quorum, rather than proposing on a timer and hoping. Sensing turns the fast-path probability from the weather term $(1 - p_{\text{miss}})^N$ into a controlled quantity bounded by gossip delay, at zero added messages (the observations ride traffic that flows anyway); the cost is measured waiting time, which we quantify in Section IX.

Theorem 3 (Liveness): Suppose $f < N/3$, each honest validator independently has an incomplete view with probability p_{miss} missing at most $k_{\text{max}} = 2$ transactions, and let δ_{sync} bound the probability that a flagged straggler fails to synchronize before the signature-phase timeout. Then the per-honest-validator probability of failing to contribute a signature is $p \leq p_{\text{miss}}e^{-n/8} + \delta_{\text{sync}}$, and consensus fails to finalize with probability at most $\exp(-2N(1/3 - p)^2)$.

Proof: An honest validator misses a signature only if falsely excluded (probability at most $e^{-n/8}$ conditioned on an incomplete view, by the threshold analysis of Section IV-C) or unsynchronized. Exclusions are independent across validators; Hoeffding over the N honest indicators against the $N/3$ failure boundary gives the bound. ■

At $n = 512$ the false-exclusion term is $p_{\text{miss}}e^{-64}$, negligible against δ_{sync} ; liveness is governed by the synchronization path, as it should be. The conference version’s false-exclusion term was an empirical 5×10^{-3} ; the analytic replacement is twenty-six orders of magnitude smaller, and exclusion is now a scheduling event rather than a participation event, since a synced straggler rejoins the signature phase.

VII. THE PROXIMA PROTOCOL

Flat protocol. The proposer disseminates the block; validators receive it with delay p_{miss} . In Phase 1 each validator sends its round digest (1024 B) and, if its state matches, the conditional commitment σ_i (96 B). The aggregator forms the median reference, clusters by the integer comparison $\|\Delta D\|^2 \leq 3n$, and synchronizes stragglers by estimate-then-decode: the digest distance sizes an IBLT whose peeling returns exactly the missing transactions, which the aggregator then pushes. With $2N/3$ commitments in hand the fast path finalizes in one round trip (Corollary 1); otherwise cluster members send signed commitments and the aggregator multicasts the aggregate plus bitmap. Every decision on the consensus path is an integer comparison.

Tree protocol. Validators form leaves of ten; leaf leaders filter by distance, synchronize stragglers, and forward the exact summary (digest sum, weight, squared-distance sum, 1036 B); internal nodes add summaries, which is exact by linearity, and the root checks the global mean. The signature phase aggregates BLS commitments up the tree. Leaves never vote, so no leaf can fail in the BFT sense: a leaf of five honest and five Byzantine validators reports the filtered mean of the honest five. Carrying speculative commitments up the tree requires per-group signer bitmaps and remains future work.

Two-syndrome synchronization. The digest is the syndrome of a random linear code over $\mathbb{Z}_q$; hard to decode, which is exactly the binding of Section V-A, but with error weight estimable from the syndrome norm (Lemma 1). IBLTs are the complementary syndrome: efficiently decodable, recovering the exact difference at communication proportional to it, but neither binding nor distance-revealing. The protocol always pairs them: estimate with the hard code, decode with the easy one, at IBLT size $1.5(\hat{d} + 6)$ cells of 16 bytes, doubling on the rare decode failure. This replaces the conference version’s Bloom filters everywhere and deletes their false-positive liveness channel outright, since IBLT peeling has no false positives.

Fork localization and partition healing. Cumulative ledger accumulators make any range digest a subtraction, so two nodes locate their fork point by binary search on prefix equality: $\log_2 H + 1$ probes of one accumulator each, with every probe also yielding $\hat{d}$ for the probed range, after which one reconciliation exchange heals the divergent suffix. During a partition the Phase 1 cloud turns bimodal; cluster centroids are exact by linearity, the inter-camp $\hat{d} = \|\mu_1 - \mu_2\|^2/n$ measures the disagreement before any repair traffic flows, and one estimate-then-decode exchange between camp representatives restores the union at cost proportional to the true divergence. Hash-based protocols observe only quorum failure in both situations. The same subtraction gives verifiable prune accounting: a node claiming to have pruned set P proves $D(\text{before}) - D(\text{after}) = D(P)$, and binding makes the claim sound.

VIII. CROSS-SHARD CONSERVATION

Sharded execution multiplies state comparison into many pairs, and hash-based tools force per-transaction coordination: two-phase commit spends four cross-shard messages per

transaction plus intra-shard consensus on each, and receipt systems [31] carry metadata proportional to volume. We instead make settlement a conservation law. Every cross-shard transaction t from shard i to shard j adds $v(t)$ (ledger domain) to i 's outbound corridor accumulator $O_{i \rightarrow j}$ when the debit applies and to j 's inbound accumulator $I_{i \rightarrow j}$ when the credit applies; each shard's quorum signs its aggregated outbound and inbound accumulators every block, and the beacon checks

$$\sum_i O_i = \sum_j I_j \quad (5)$$

over a settlement window W : double-entry bookkeeping in vector space, at $O(S)$ kilobyte-scale signed objects per block regardless of volume.

Theorem 4 (Conservation soundness): If the invariant holds over quorum-signed accumulators and the accumulator is collision-binding, then the multiset of applied debits equals the multiset of applied credits except with negligible probability; any minted, lost, or replayed cross-shard effect requires forging a quorum signature or finding a digest collision.

Proof: The invariant asserts $D(\text{debits}) = D(\text{credits})$ in the ledger domain. If the underlying multisets differ, the pair is a collision (Section V-A); if a shard signs an accumulator inconsistent with its applied transactions, safety of its intra-shard quorum is violated. Value conservation follows from multiset conservation plus intra-shard execution validity, since amounts are bound inside the hashed transactions; we deliberately do not weight vectors by amounts, which would weaken the SIS instance. ■

Proposition 1 (Localization by group testing): On invariant failure, the beacon bisects the corridor set with digests aggregated over halves, recursing into imbalanced halves. Bisection over a linear measurement is exact, and f faulty corridors among C are localized in at most $2f(\log_2(C/f) + 2)$ probes, each one signed kilobyte-scale object.

Two refinements make the mechanism operational. First, the per-corridor *pending digest* $P_{ij} = O_{i \rightarrow j} - I_{i \rightarrow j}$ is not merely an alarm signal: it is the ledger digest of the in-flight transaction set on that corridor, so its norm meters in-flight volume continuously, and the invariant is enforced as “ P_{ij} decays to zero within W blocks,” with aging alarms separating ordinary settlement lag from faults, since credits necessarily trail debits by network latency. Second, each corridor can carry the IBLT syndrome of the same in-flight set alongside P_{ij} (both objects are linear), so an alarmed corridor decodes exactly its dangling transactions, dropped credits and minted credits alike, from the source's applied-debit set against the destination's applied-credit set, with no further exchange.

The semantics deserve a plain statement: this is detection and settlement verification, not locking. Digests order nothing; the protocol pairs the invariant with optimistic execution and a W -block revert window, or credits at the destination only on digest match, which is receipt-equivalent in latency but constant-size in verification metadata. The contribution is the verification and fraud-localization cost, and the fair baseline comparison is per-block verification metadata against batched two-phase commit and batched receipts, which we report in Section IX. A beacon may additionally maintain the single running digest $G \leftarrow \sum_i \Delta_i$ per block, one 2KB object summarizing the entire

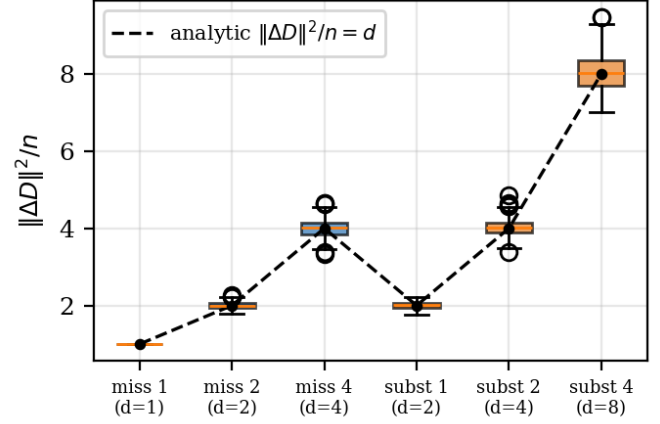


Fig. 1. Squared digest distance over n against true symmetric difference, for missing (blue) and substituted (orange) transactions, 400 trials per case at $n = 512$. Boxes collapse onto the analytic line; $d = 1$ has zero variance.

system's applied history, updatable by addition and prunable by subtraction with verifiable prune accounting, none of which a Merkle root supports.

IX. EVALUATION

All experiments run in the discrete-event simulator of the public artifact, which counts every message with identical byte constants across protocols; BLS operations use mocks preserving wire sizes, cross-validated against real BLS12-381 on small runs. The suite of 41 correctness checks, including the exact test vectors of Section IV, passes on every run reported here. Baselines are HotStuff [3], HotStuff-2 [8], PBFT [2], and a message-count model of Kauri [9] (fanout 10) that credits its tree dissemination but not its pipelining and does not charge its reconfiguration on relay failure, a deliberately generous simplification we return to below. Unless stated, N splits 70/30 honest/Byzantine, $p_{\text{miss}} = 0.37$ with $k_{\text{max}} = 2$, and blocks carry 21 transactions.

A. The Estimator Behaves as Proved

Fig. 1 shows $\|\Delta D\|^2/n$ against ground truth for missing and substituted transactions: samples collapse onto the analytic line $\hat{d} = d$, the $d = 1$ case is exactly n with zero observed variance across 400 trials, and substitutions read their full symmetric difference, closing the conference version's blind spot. Fig. 2 plots the exact small- d laws against $\tau^2 = 3n$: $d \leq 2$ sits cleanly below threshold (0 exclusions in 500 validation trials, against the e^{-64} bound), $d = 4$ cleanly above, and $d = 3$ straddles it, resolved by synchronization as designed.

B. Reference Robustness

Fig. 3 mounts a coordinated drag attack: all Byzantine validators submit one extreme digest and the aggregator is itself Byzantine. With the conference version's aggregator-chosen reference the adversary excludes 100% of honest validators at every corruption level, a total liveness failure available to a single corrupted role. The coordinate-wise median reference excludes no honest validator up to the 45% corruption we tested, matching its breakdown guarantee.

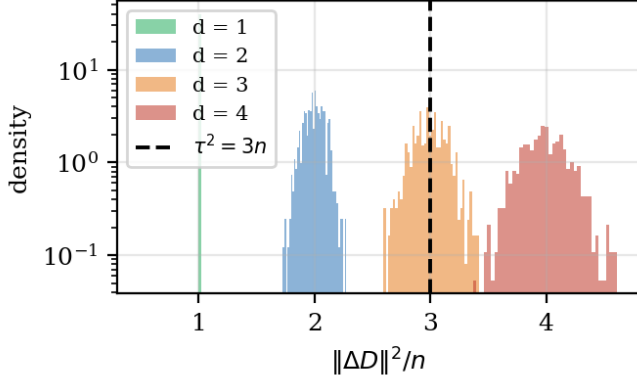


Fig. 2. Distributions of $\|\Delta D\|^2/n$ at $d = 1, \dots, 4$ against the closed-form threshold $\tau^2 = 3n$ (log scale). $d = 1$ is a point mass at exactly 1; honest stragglers ($d \leq 2$) never cross the threshold; $d = 3$ is the designed ambiguity region.

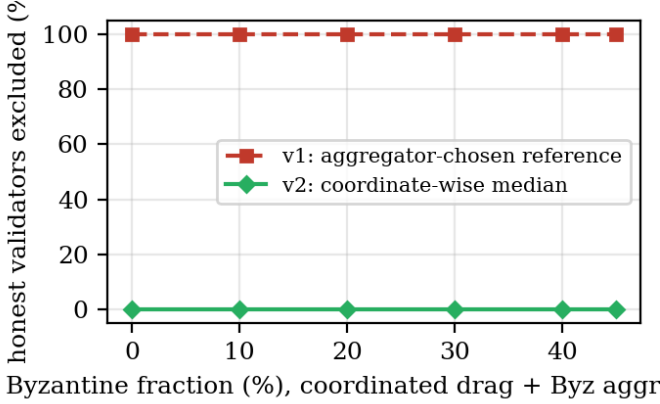


Fig. 3. Honest exclusion under coordinated drag with a Byzantine aggregator, $N = 90$, eight trials per point. The aggregator-chosen reference (v1) admits total exclusion; the coordinate-wise median (v2) admits none through 45% corruption.

C. Fast Path: Griefing and Sensing

Fig. 4 (left) is the griefing experiment: mimicking Byzantine validators sit inside the cluster and perturb its variance. The variance trigger of the conference version drops from 100% to 0% fast-path engagement at a single such adversary and never recovers; the commitment-counting trigger is indifferent through eight, as honest-sufficiency predicts. Fig. 4 (right) evaluates readiness sensing at increasing initial miss rates with a gossip fill probability of 0.5 per tick and an 8-tick cap: fixed-interval proposal collapses to a 6% fast-path rate at $p_{\text{miss}} = 0.2$ and to zero beyond 0.4, while the sensing proposer waits a measured mean of 1.3 to 3.4 ticks and engages the fast path in every trial through $p_{\text{miss}} = 0.8$. Sensing spends bounded, observable delay to buy a round; the exchange rate is the gossip fill time, not a protocol constant.

D. Synchronization Cost

Fig. 5 compares straggler synchronization sketches at local set size 500. The Bloom filter of the conference version costs a flat 599 bytes because it encodes the whole set, and carries

TABLE II
MESSAGES AND KB PER BLOCK (30% BYZANTINE, $p_{\text{miss}} = 0.37$).

protocol	$N = 1000$		$N = 3000$	
	msgs	KB	msgs	KB
Proxima Tree	3,286	1,506	9,818	5,103
Proxima Flat	3,644	1,415	10,884	4,742
HotStuff	6,518	1,322	19,554	6,164
HotStuff-2	5,518	1,195	16,554	5,782
Kauri (msg model)	7,511	1,105	22,547	4,297
PBFT	1,999,000	249,875	17,997,000	2,249,625

a false-positive channel besides; the IBLT costs 256 bytes up to $d = 4$ and grows linearly in the true difference with exact decode (every trial decoded in one round at the estimator-supplied size). Below $d \approx 16$ the exact sketch is also the cheaper one; beyond, it pays only for what actually differs.

E. Conservation

Fig. 6 (left) plots per-block verification metadata at $S = 16$ shards: batched receipts and batched two-phase commit grow with volume, conservation is flat at 67 KB, and the crossover sits at roughly 2,100 cross-shard transactions per block, below which receipts are cheaper and above which conservation wins unboundedly. Fig. 6 (right) confirms Proposition 1: measured localization probes at $C = 64$ corridors track $2f(\log_2(C/f) + 2)$ and localization was exact in every run. The end-to-end fault run of Section VIII (16 shards, 14,400 transactions, 9 injected faults across two corridors, one of them a minted credit) detected the violation with a dangling-effect estimate of 8.9 against the true 9, named exactly the two faulty corridors both by aging alarms and by 15 bisection probes, and the corridor syndrome decode recovered every dropped and minted transaction exactly.

F. Scaling Against Modern Baselines

Fig. 7 and Table II give message and bandwidth scaling to $N = 3000$ with 30% Byzantine and 37% partial observation. On messages, Proxima Tree runs $1.99\times$ below HotStuff and $1.69\times$ below HotStuff-2 at $N = 3000$, with the Kauri model above flat HotStuff since tree dissemination adds edges without removing voting; the conference version's headline held only against three-phase HotStuff, and this is the corrected claim. On bandwidth the picture inverts and is worth stating carefully: kilobyte digests make Proxima 1.4 to $1.6\times$ heavier than HotStuff-2 at $N \leq 300$, the gap closes through $N = 1000$, and beyond $N \approx 2000$ Proxima is lighter, because certificate bitmaps broadcast to all validators cost the hash-based protocols $\Theta(N^2/8)$ bytes while digests cost $\Theta(N)$. Deployments below the crossover can take the light profile (256-byte digests at 12.5% estimator error, Table I); we report the default profile throughout.

G. Fork Localization and Partition Healing

Fig. 8 measures the accumulator machinery: fork localization uses exactly $\log_2 H + 1$ probes from $H = 64$ to $H = 4096$ (7 to 13 probes, each one 2 KB accumulator), with every probe

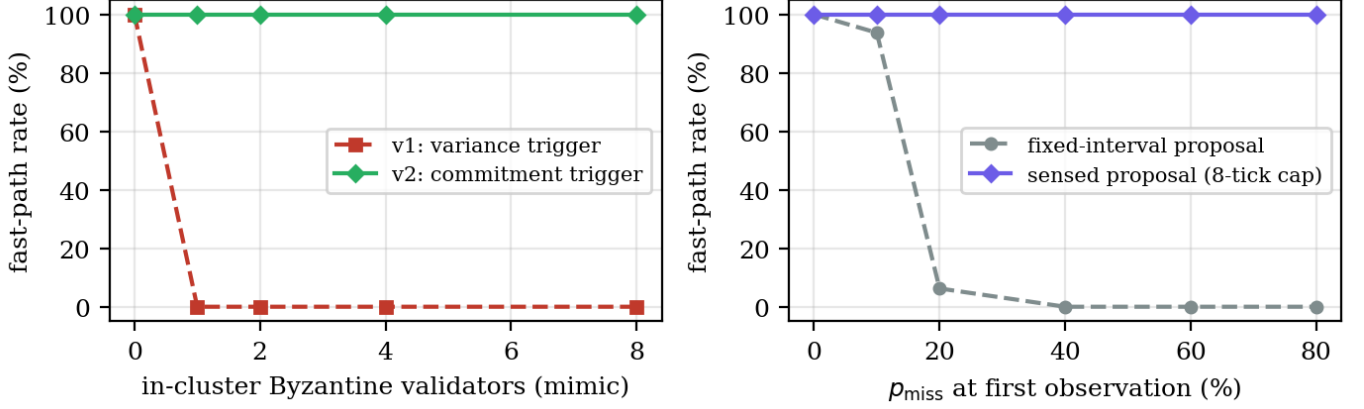


Fig. 4. Left: fast-path rate against in-cluster mimicking Byzantine validators ($N = 60$ honest, complete views); the variance trigger dies at one adversary, the commitment trigger is unaffected. Right: fast-path rate against initial p_{miss} for fixed-interval versus sensed proposal (fill probability 0.5 per tick, cap 8 ticks, mean measured delay 0 to 3.4 ticks).

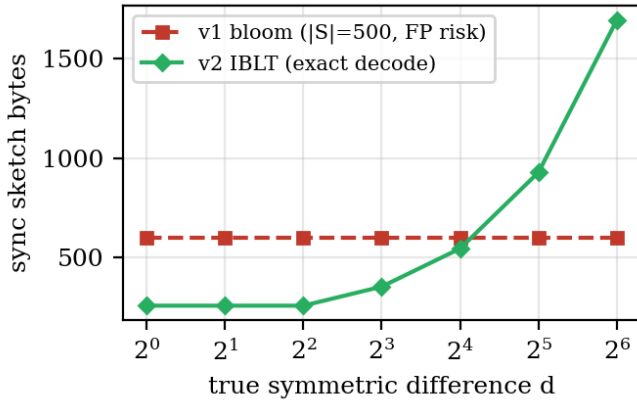


Fig. 5. Sync sketch bytes against true difference at set size 500. The Bloom filter scales with the set and admits false positives; the IBLT scales with the difference and decodes exactly.

also reporting how much diverged; partition healing at 600-transaction state recovers the exact union at sketch-plus-push cost tracking the 200-bytes-per-divergent-transaction floor, with the pre-repair estimate $\hat{d}$ within its predicted error of the true divergence throughout.

H. What the Simulation Does Not Show

Three honest limits bound these results. The Kauri numbers are a message-count model that credits neither pipelining, Kauri’s central contribution to throughput, nor charges reconfiguration under relay failure; a faithful comparison needs Kauri’s own harness. Latency is not reported at all: our earlier projections mixed measured constants with modeled round trips, and we now regard a multi-region deployment as the only honest source of latency claims (Section X). And $p_{\text{miss}} = 0.37$ remains an internal measurement; the sensing results deliberately sweep it as a free parameter rather than defending the point value.

X. LIMITATIONS AND FUTURE WORK

Concrete cryptanalysis. Binding rests on a reduction shape to a ± 1 -matrix SIS instance plus first-cut Wagner and lattice estimates and the inherited LtHash analysis; a dedicated cryptanalysis at exactly our parameters ($n = 512$, $q \in \{2^{16}, 2^{32}\}$, block-bounded weight) is open, and until it exists we claim no numeric security level.

Deployment. All numbers are simulation; a 50-to-200 node deployment across three regions would convert message counts into latency and throughput measurements and let us measure p_{miss} rather than assume it.

Adaptive adversaries. The model is static corruption. An adversary who corrupts after observing tree assignments or sensing behavior requires VRF-based assignment with epoch rotation, described but not evaluated here; note that Theorem 2 is indifferent to this choice, since it holds against whatever adversary class Π tolerates.

Tree fast path. Carrying speculative commitments up the tree requires per-group signer bitmaps; the flat fast path does not extend without them.

Value semantics. Conservation verifies multiset settlement, not execution ordering; composing the invariant with an execution layer that needs atomicity across more than two shards is future work.

XI. CONCLUSION

Collision-resistant hashing gives consensus protocols a one-bit view of agreement, and the costs of that single bit run through the BFT literature: mandatory synchronization, unmeasurable agreement, oversized committees, per-transaction cross-shard coordination. The repair is not a better hash but a division of labor that we prove is forced: estimation objects cannot bind and binding objects cannot estimate, yet one zero-mean additive multiset hash provides both faces, an unbiased symmetric-difference estimator on its geometry and a SIS-hard homomorphic accumulator on its equality. Around that object we rebuilt a BFT protocol in which every optimistic mechanism fires on honest-sufficient conditions and every

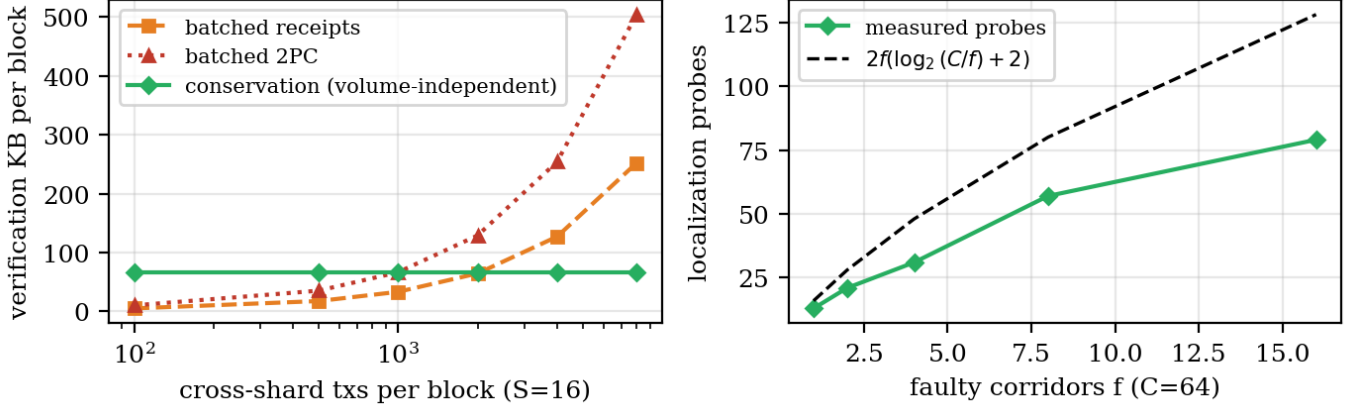


Fig. 6. Left: per-block verification metadata at $S = 16$; conservation is volume-independent, with the receipts crossover near 2,100 cross-shard transactions per block. Right: localization probes against faulty-corridor count at $C = 64$, with the analytic budget.

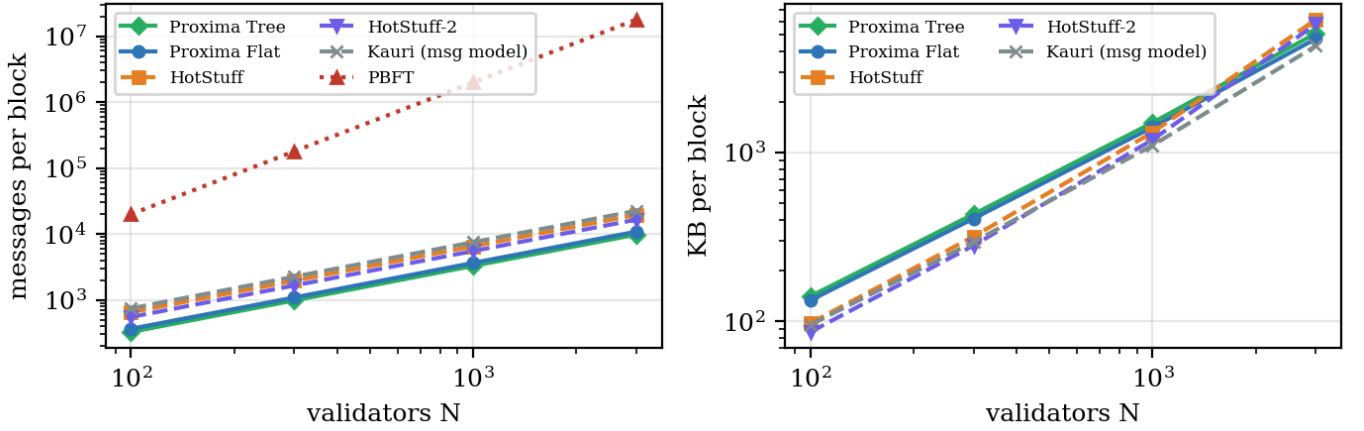


Fig. 7. Message (left) and bandwidth (right) scaling, log-log. Proxima leads on messages throughout; on bandwidth the kilobyte digests cost a premium at small N that inverts beyond $N \approx 2000$ as $\Theta(N^2/8)$ certificate bitmaps overtake $\Theta(N)$ digests.

estimate is repaired by an exact decoder, and we extended the same algebra to sharded settlement as a conservation law with logarithmic fraud localization. The composition theorem guarantees all of it costs nothing in safety; the measurements say it pays in messages, in robustness to griefing and reference manipulation, and, at scale, in bandwidth.

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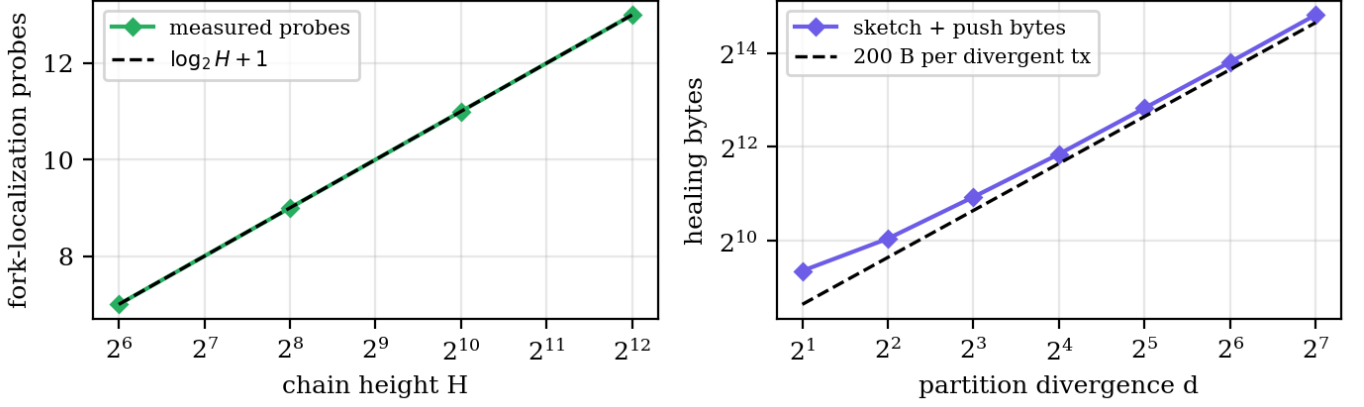


Fig. 8. Left: fork-localization probes against chain height, matching $\log_2 H + 1$ exactly. Right: partition-healing bytes against divergence, tracking the per-transaction push floor.

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